

Probabilistic reconstruction of charge stability diagrams from sparse ray measurements

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Spin qubits in gate-defined quantum dots are a leading candidate for scalable quantum computation, and operating them requires the charge occupation of each dot to be known. This occupation is read from the charge stability diagram (CSD), which is conventionally acquired as a dense two-dimensional scan whose acquisition time grows quadratically with resolution. Recent generative models reconstruct the full sensor image from a sparse scan, and the transition lines are then extracted from that image with classical edge and ridge detectors. These models are trained on measured data whose reference lines come from the same detectors and inherit their errors. We instead formulate the task directly as the probabilistic reconstruction of the transition lines themselves. A U-Net, trained entirely on a constant-capacitance simulator whose transition lines are exact by construction, takes a few one-dimensional voltage sweeps (rays) through the gate-voltage plane and returns, for every pixel, the probability that a transition line passes through it. This map is binarised with a threshold selected on validation data, and the result is scored with a tolerance-based F1 score that does not penalise one-pixel displacements of otherwise correct lines. With eight rays sampling 4.7 % of the voltage plane, the transition lines of 50 held-out simulated devices are recovered with $F1 = 0.82$ at a one-pixel tolerance. Application to measured devices remains to be demonstrated.